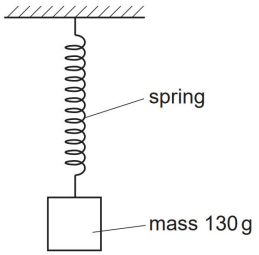
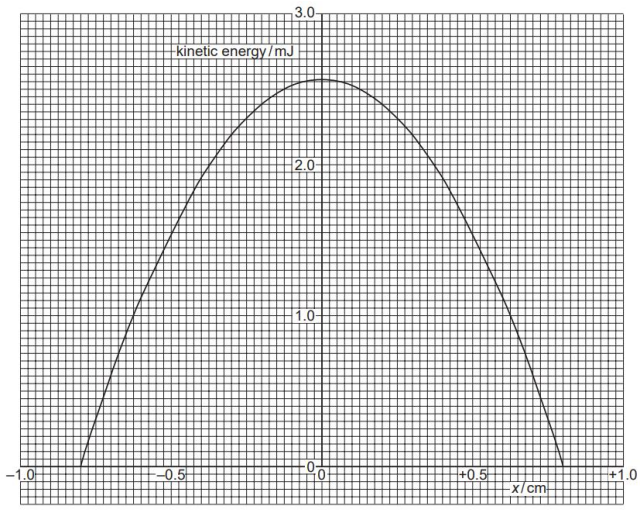
17.2 Exercises - Energy in S.H.M.

1. A spring is hung from a fixed point. A mass of 130 g is hung from the free end of the spring, as shown in Fig.





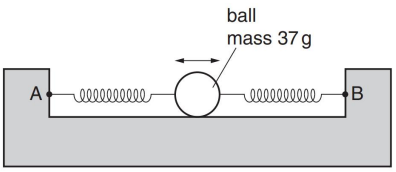
The mass is pulled downwards from its equilibrium position through a small distance d and is released. The mass undergoes simple harmonic motion. Fig. shows the variation with displacement x from the equilibrium position of the kinetic energy of the mass. Use Fig. to

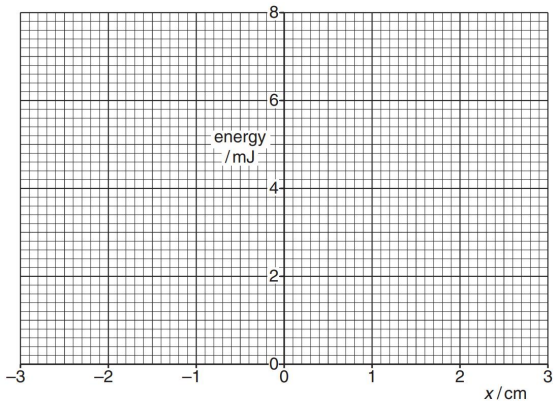
a. determine the distance d through which the mass was displaced initially,

d = ... cm [1]

b. show that the frequency of oscillation of the mass is approximately 4.0 Hz. [6]

2. A ball of mass 37 g is held between two fixed points A and B by two stretched helical springs, as shown in Fig.





The ball oscillates along the line AS with simple harmonic motion of frequency 3.5 Hz and amplitude 2.8 cm.

a. Show that the total energy of the oscillations is 7.0 mJ. [2]

b. At two points in the oscillation of the ball, its kinetic energy is equal to the potential energy stored in the springs. Calculate the magnitude of the displacement at which this occurs.

displacement = ... cm[3]

c. On the axes of Fig. and using your answers in (a) and (b), sketch a graph to show the variation with displacement x of

i. the total energy of the system (label this line T), [1]

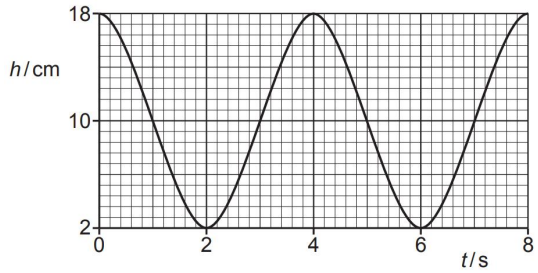
ii. the kinetic energy of the ball (label this line K), [2]

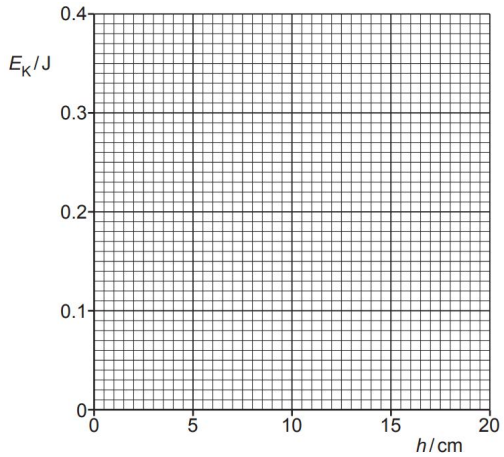
iii. the potential energy stored in the springs (label this line P). [2]

d. The arrangement in Fig. is now rotated through 90° so that the line AB is vertical and the ball oscillates in a vertical plane. Suggest one form of energy, other than those in (c), that must be taken into consideration when plotting new graphs to show energy changes with displacement.

.....[1]

3. Fig. shows the variation with time t of the height h above the ground of an object of mass 36kg that is undergoing vertical simple harmonic motion.





a. For the oscillations of the object:

i. determine the amplitude x_0, in cm

x_0 = ...cm [1]

ii. show that the angular frequency ω is 1.6 $rads^{-1}$. [2]

iii. determine the total energy E.

E = ...J [3]

b. On Fig. below, sketch the variation with h of the kinetic energy E_K of the object. [4]